

YSP SALBUTAMOL INJECTION 0.5mg/ml

Ingredient(s):

Each ml contains:

Salbutamol Sulfate 0.6mg
(eq. to Salbutamol 0.5mg)

Pharmacodynamic:

Salbutamol is a selective β_2 adrenoceptor agonist. At therapeutic doses it acts on the β_2 adrenoceptors in the bronchi and uterus, with little or no action on the β_1 adrenoceptors of the heart. It is suitable for the management of an asthmatic attack, and for uncomplicated premature labour, under the direction of a physician.

Pharmacokinetic:

Salbutamol administered intravenously has a half-life of 4 to 6 hours and is cleared partly renally and partly by metabolism to the inactive 4'-O-sulphate (phenolic sulphate) which is also excreted primarily in the urine. The faeces are a minor route of excretion. The majority of a dose of salbutamol given intravenously, orally or by inhalation is excreted within 72 hours. Salbutamol is bound to plasma proteins to the extent of 10%.

Indication(s):

1. Relief of severe bronchospasm associated with asthma or bronchitis and for the treatment of status asthmaticus.
2. Management of uncomplicated premature labour in the last trimester of pregnancy.

Dosage and Administration(s):

Adults

In severe Bronchospasm and Status Asthmaticus:

Subcutaneous Route

500 micrograms (8 micrograms /kg bodyweight) and repeated every four hours as required.

Intramuscular Route

500 micrograms (8 micrograms /kg bodyweight) and repeated every four hours as required.

Intravenous Route - Injection

250 micrograms (4 micrograms /kg bodyweight) injected slowly. If necessary the dose may be repeated.

Intravenous Route – Infusion

A suitable solution for infusion may be repeated by diluting 5mL of Salbutamol Solution for IV infusion in 250mL of an infusion solution eg. Sodium Chloride and Glucose Solution to provide a Salbutamol dose of 10 mcg/mL of solution.

Adults: Infusion rate of 3-20 mcg/min are generally adequate. Higher doses have been used with success in patients with respiratory failure.

In the Management of Premature Labour:

The infusion should be administered as early as possible after the diagnosis of premature labour, and after evaluation of the patient to rule out contraindications to the use of salbutamol.

Infusion rates of 10-50 mcg/min are generally adequate to control uterine contractions, but greater or lesser infusion rates may be required according to the strength and frequency of contractions. A starting dose of 10 mcg/min is recommended, increasing the rate at 10 min intervals until there is evidence of patient response shown by diminution in strength, frequency or duration of contractions. Thereafter, the infusion rate may be increased slowly until contractions cease. The maternal pulse rate should be monitored and the infusion rate adjusted to avoid excessive maternal heart rates (above 120 beats/min).

Once uterine contractions have ceased the infusion rate should be maintained at the same level for one hour and then reduced by 50% decrements at 6 hourly intervals. The infusion should be stopped if labour progresses despite treatment. Continuation of Salbutamol Injection for more than 48 hours is not recommended.

As an alternative procedure or to counteract inadvertent overdosage with oxytocic drugs, Salbutamol injection may be administered as a single injection by the IV or IM routes. The usual recommended dose is 100-250 mcg of Salbutamol. The dose may be repeated according to the response of the patient.

Children

At present there is insufficient evidence to recommend a dosage regimen for routine use in children.

Mode of Administration:

Subcutaneous, intramuscular and intravenous.

Contraindication(s):

Salbutamol preparations are contraindicated in patients with a history of hypersensitivity to any of their components.

Although intravenous salbutamol and occasionally salbutamol tablets are used in the management of premature labour, uncomplicated by conditions such as placenta praevia, ante-partum haemorrhage or toxemia of pregnancy, salbutamol preparations should not be used for threatened abortion.

Precaution(s) / Warning(s):

1. The management of asthma should normally follow a stepwise programme, and patient response should be monitored clinically and by lung function tests.
2. Increasing use of short-acting inhaled β_2 agonists to control symptoms indicates deterioration of asthma control. Under these conditions, the patient's therapy plan should be reassessed. Sudden and progressive deterioration in asthma control is potentially life threatening and consideration should be given to starting or increasing corticosteroid therapy. In patients considered at risk, daily peak flow monitoring may be instituted.
3. The use of salbutamol parenteral preparations in the treatment of severe bronchospasm or status asthmaticus does not obviate the requirement for glucocorticoid steroid therapy as appropriate.
4. When practicable, administration of oxygen concurrently with parenteral salbutamol is recommended, particularly when it is given by intravenous infusion to hypoxic patients.
5. In common with other β -adrenoceptor agonists, salbutamol can induce reversible metabolic changes such as reversible hypokalaemia and increased blood glucose levels. The diabetic patient may be unable to compensate for this and the development of ketoacidosis has been reported. Concurrent administration of corticosteroids can exaggerate this effect.
6. Potentially serious hypokalaemia may result from β_2 agonist therapy mainly from parenteral and nebulised administration. Particular caution is advised in acute severe asthma as this effect may be potentiated by concomitant treatment with xanthine derivatives, steroids, diuretics and by hypoxia. It is recommended that serum potassium levels are monitored in such situations.
7. Lactic acidosis has been reported very rarely in association with high therapeutic doses of intravenous and nebulised short-acting beta-agonist therapy, mainly in patients being treated for an acute asthma exacerbation (see Adverse Reaction section). Increase in lactate levels may lead to dyspnoea and compensatory hyperventilation, which could be misinterpreted as a sign of asthma treatment failure and lead to inappropriate intensification of short-acting beta-agonist treatment. It is therefore recommended that patients are monitored for the development of elevated serum lactate and consequent metabolic acidosis in this setting.
8. Salbutamol should be administered cautiously to patients with thyrotoxicosis.
9. Tocolysis: Serious adverse reactions including death have been reported of salbutamol to women in labor. In the mother, these include increased heart rate, transient hyperglycaemia, hypokalaemia, cardiac arrhythmias, pulmonary oedema and myocardial ischaemia. Increased fetal heart rate and neonatal hypoglycaemia may occur as a result of maternal administration.

10. As maternal pulmonary oedema and myocardial ischaemia have been reported during or following premature labour in patients receiving β_2 – agonists, careful attention should be given to fluid balance and cardio-respiratory function, including ECG monitoring. If signs of pulmonary oedema and myocardial ischaemia develop, discontinuation of treatment should be considered.
11. Due to the risk of pulmonary oedema and myocardial ischaemia that has been observed during the use of beta2-agonists in the treatment of premature labour, before these products are given to any patient with known heart disease, an adequate assessment of the patients's cardiovascular status should be made by a physician experienced in cardiology.
12. Cautious use of salbutamol injections is required in pregnant patients when it is given for relief of bronchospasm so as to avoid interference with uterine contractibility. During IV infusion of salbutamol, the maternal pulse should be monitored and not normally allowed to exceed a steady rate of 140 beats per minute.
13. Effect on ability to drive and use machines – none known.

Interaction with Other Medicaments:

1. Salbutamol and non-selective beta-blocking drugs, such as propranolol, should not usually be prescribed together.
2. Concomitant use of salbutamol and tricyclic antidepressants or monoamine oxidase inhibitors may cause a potentiation of the vascular effects of salbutamol. Salbutamol is not contra-indicated in patients under treatment with monoamine oxidase inhibitors (MAOIs).

Pregnancy and Lactation:

Administration of drugs during pregnancy should only be considered if the expected benefit to the mother is greater than any possible risk to the foetus.

As salbutamol is probably secreted in breast milk its use in nursing mothers is not recommended unless the expected benefits outweigh any potential risk. It is not known whether salbutamol in breast milk has a harmful effect on the neonate.

Side Effect(s):

Immune system disorders

Hypersensitivity reactions including angioedema, urticaria, bronchospasm, hypotension and collapse

Metabolism and nutrition disorders

Hypokalaemia, Lactic acidosis

Nervous system disorders

Tremor, headache, hyperactivity

Cardiac disorders

Tachycardia, palpitations, myocardial ischaemia (in the management of pre-term labour with salbutamol injection/solution for infusion), cardiac arrhythmias including atrial fibrillation, supraventricular tachycardia and extrasystoles.

Vascular disorders

Peripheral vasodilatation.

Respiratory, thoracic and mediastinal disorders

Pulmonary oedema.

Gastrointestinal disorders

Nausea, vomiting

Musculoskeletal and connective tissue disorders

Muscle cramps

Injury, poisoning and procedural complications

Slight pain or stinging on i.m. use of undiluted injection.

Symptoms and Treatment for Overdosage, and Antidote(s):

Overdosage symptoms are those of excessive β -stimulation, e.g. seizures, angina, hypertension or hypotension, tachycardia with rates up to 200 beats/min, arrhythmias, nervousness, headache, tremor, dry mouth, palpitation, nausea, dizziness, fatigue and insomnia. Hypokalaemia may occur following overdose with salbutamol. Serum potassium levels should be monitored.

Treatment consists of discontinuation of salbutamol together with appropriate symptomatic therapy. Administer a cardioselective β -adrenergic blocker (e.g. acebutalol, atenolol, metoprolol), if necessary for cardiac arrhythmias. However, β -adrenergic blocker should be used with caution because it could induce severe bronchospasm.

Shelf-Life:

3 years

Storage Condition(s):

Store at temperature below 30°C. Protect from light and moisture.

Compatibilities:

The following infusion fluids are compatible with YSP Salbutamol Injection 0.5mg/ml:

5% w/v Glucose Solution

0.9% w/v Sodium Chloride Solution

0.18% Sodium Chloride and 4% w/v Glucose Solution

Instruction for Use:

- i) The prepared infusion solution should be made up immediately before use.
- ii) The diluents are compatible to be administered via Intravenous, Intramuscular & Subcutaneous route.
- iii) This product is physically and chemically compatible for up to 24 hours for following diluents:
 - 5% Glucose Solution or;
 - 0.9% Sodium Chloride Solution or
 - 0.18% Sodium Chloride and 4% w/v Glucose Solution.
- iv) 1ml of the product can be diluted into 10 to 50ml of diluents.

Product Description:

A clear and colorless to pale yellow color solution

Dosage Forms and Packaging available:

1ml x 10, 1ml x 50 and 1ml x 100 clear ampoules.



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